

OPPONG KWABENA ADUTWUM

COMPUTER SCIENCE UNDERGRADUATE

Applied ML · Robotics · Embedded Intelligence · Full-Stack Development

CONTACT

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github.com/Oppong-py
[oppoing-py.github.io](https://github.com/oppoing-py)

TECHNICAL SKILLS

Programming Languages

- Python
- C++

Data & Analytics

- NumPy / pandas
- Matplotlib / Seaborn
- SQL
- Jupyter Notebooks
- Preprocessing & EDA

AI & ML Tools

- Scikit-learn
- Groq AI API
- Claude Code / Claude Cowork

Core Knowledge

- Machine Learning Fundamentals
- Transfer Learning
- Cross-validation & Leakage
- PID Control / Robotics
- Data Structures & Algorithms
- OSI & TCP/IP Networking

Web & Backend

- React Native / Expo

Dev Tools & Design

- Git / GitHub
- VS Code
- Figma
- Linux (Kali)

Learning now

- PyTorch
- TensorFlow
- Hugging Face
- OpenCV
- Django
- REST API Design

LANGUAGES

- English — Native
- Twi — Native
- French — Intermediate

CERTIFICATIONS

- Certificate of Participation — Ghana Data Science Regional Hackathon (Sep 2026)
- Hedera Certified Foundation (Apr 2026)
- Hashgraph Developer Course — Hedera (Jan 2026)
- WASSCE General Science — Prempeh College (Sep 2025)
- IPPF Competitor — Brewer Foundation & NYU (2024-25)

GITHUB

- [gds-hackathon-2026-building-age](https://github.com/oppoing-py/gds-hackathon-2026-building-age)
- [ml-specialization-coursera](https://github.com/oppoing-py/ml-specialization-coursera)
- [oppoing-py.github.io](https://github.com/oppoing-py)

MEMBERSHIPS

- Prempeh College Robotics Club (2023–2025)
- AmaliTech Coding Club (2025–Present)
- Microsoft User Group — KNUST (2025–Present)
- Claude Builder Club — KNUST (2025–Present)

◆ PROFESSIONAL SUMMARY

Second-year Computer Science student at Kwame Nkrumah University of Science and Technology. Started with robotics, went through network security, and has landed on applied machine learning. National Robotics Champion (Robofest Ghana 2025) and Regional Cybersecurity Champion (2024). Both came from the same habit: watch what a system does, change one thing, watch again.

Most recent work is a four-day satellite-imagery problem at the Ghana Data Science Regional Hackathon 2026: estimating building age from 40 years of Landsat readings, trained on Madrid and transferred to Amsterdam. What came out of it was finding the supplied evaluation inflating every score by 0.047 macro-F1 through spatial leakage, and that the feature set scoring best on the training city was the worst one to carry across. Both findings are public, with the code.

◆ EDUCATION

Kwame Nkrumah University of Science & Technology (KNUST)

B.Sc. Computer Science | Kumasi, Ghana | Expected 2029

- Relevant coursework: Data Structures, Algorithms, Discrete Mathematics, Linear Algebra, Probability & Statistics, C++ Programming, Networking & Internet (OSI/TCP/IP, IPv4, subnetting), Calculus I.

Prempeh College (Senior High School)

WASSCE — General Science | Kumasi, Ghana | Graduated September 2025

- Left with strong marks in Science, Mathematics and Elective ICT.
- Founded and led the school's Cybersecurity Club (2023–2025), running sessions on network security, ethical hacking and CTF challenges, and mentoring 7 junior students.
- Competed nationally with the Robotics Club and won Robofest Ghana 2025.

◆ EXPERIENCE

Machine Learning Trainee — ZapTek Research and Innovation Hub

Remote | May 2026

- Completed a 2-week introductory ML programme covering data collection, preprocessing, model training and evaluation; currently in the second cohort.
- Cleaned datasets using pandas and NumPy, handling missing values and feature scaling.
- Trained classification models with Scikit-learn and compared performance using accuracy, precision, recall and F1-score.

Online Safety Analyst & Volunteer — Solutions for Life Initiative Ghana (SFLIG)

Ghana | Oct 2022 – Present

- Joined as a volunteer in October 2022 on community support projects, donating food, sanitary items and clothing to rural communities.
- Promoted to Online Safety Analyst in August 2023. Manages the organisation's social media and website with a focus on internet security.
- Runs speaking engagements on staying safe online with girls from orphanages and rural communities.

Co-Founder & Software Developer — Arethos Tech Initiatives

Kumasi, Ghana | 2026 – Present

- Co-founding and leading development of Arethos, an offline-first mental wellness app for KNUST students. Managing the technical roadmap and coordinating a three-person team.
- Engineered the reflection engine on Groq for fast inference, and a custom haptic layer calibrated across nine iterations on Android hardware.
- Won 2nd Place at the ACES CodeFest 2026 UI/UX Challenge with a team of five.

◆ PROJECTS

Building Age Classification from Satellite Imagery

Python · scikit-learn · Random Forest · scipy.spatial · Landsat time series

- Estimated construction era for 102,255 labelled pixels from 40 years of Landsat readings; trained on Madrid, transferred to Amsterdam with as few as five labelled examples per era.
- Found the supplied evaluation inflating every score by 0.047 macro-F1: a satellite pixel is smaller than a building, so a random split put parts of the same roof in both train and test. Measured with a kDTree adjacency check, 82.4% of test pixels were touching a training pixel. Replaced it with blocked cross-validation and a buffer.
- Showed that feature sets rank in the opposite order on the target city, so choosing a model by its best home score picks the wrong one.
- Four days at KNUST with Kelly Nyarko, Hughes Tsikata and Antwi Yaa. The team did not place. Code and both write-ups are public: github.com/Oppong-py/gds-hackathon-2026-building-age

ACES KNUST Website Redesign

Figma · Mobile-first UI/UX · 2nd Place, CodeFest 2026

- Redesigned all nine pages of the ACES KNUST site for mobile, homepage through marketplace and course resources, as a high-fidelity clickable prototype.
- Built with real-time collaboration across a team of five: Emmanuel Oduro-Okyireh, Richmond Baako, Kenan Tetteh, Caleb Obeng and myself. Focused on visual hierarchy, touch-friendly navigation and real empty states.

Arethos — Mental Wellness App (In Development)

React Native · Expo SDK 54 · SQLite · Groq AI

- Cross-platform app for KNUST students: an AI-assisted journal, a mood tracker logging sleep and energy, and a reflections tab that surfaces behavioural patterns in plain language.
- Fully offline-first on SQLite; no user data leaves the device.
- Diagnosed and fixed three instructive bugs: React Native `zIndex` hit-testing between non-sibling elements, a stale closure in a PanResponder gesture handler, and nine haptic iterations ending in native `Vibration.vibrate()` because expo-haptics fires only single pulses on Android.
- Documented across 16 chapters covering every architectural decision. Repository private until release; the engineering write-up is public.

INTERESTS

- Mathematics
- Machine Learning & AI
- Robotics & Embedded Systems
- UI/UX & Visual Design
- Movies & Video Games
- Sports (Football)

Autonomous Line-Following Robot

LEGO MINDSTORM EV3 · PID Control Algorithm

- Designed and hand-tuned a PID control algorithm for autonomous line-following; National Champion at Robofest Ghana 2025.
- Tuned the sensor-motor feedback loop by iteration: run, watch the oscillation, adjust, run again, under strict competition time limits.

Solar-Powered Automated Water Dispenser

Arduino · Ultrasonic Sensors · C++ · Solar

- Dual ultrasonic sensors detect a container and open a solenoid valve; fill level is monitored in real time and the valve closes when the container is full or taken away.
- Debounce logic prevents false triggers, with a buzzer for audio feedback and LED status indicators. Runs off-grid on solar for deployment where mains power is unreliable.

Class Attendance Monitoring System

React Native · Django · PostgreSQL · Bluetooth Low Energy

- Built a full-stack biometric attendance platform and tested it with one class of 50 students.
- The engineering problem was concurrency: several phones checking in at the same moment, and the duplicate and interleaved writes that come with it. Handled in a Django REST API backed by PostgreSQL.

MACHINE LEARNING WORK PLANNED

Student Performance Prediction

NumPy, from scratch

- Logistic regression written by hand with no scikit-learn, building the gradient descent loop directly so the mathematics stops being a black box.

Crop Disease Detector

Convolutional networks · applied computer vision

- Train a CNN on labelled plant images to identify disease from a single leaf photo, served behind a web app that returns a diagnosis with treatment guidance.

AI Study Assistant

Retrieval-augmented generation

- A chatbot that ingests textbooks and lecture notes and answers student questions using vector search, considered prompt design and safety guardrails.

Energy Demand Forecasting

Regression · time series

- Predict energy demand from historical consumption and weather, deployed as an interactive dashboard.

HACKATHONS & COMPETITIONS

Ghana Data Science Regional Hackathon 2026 — Participant — Machine Learning (Sept 2026)

- Organised by the AstroXMat Research Laboratory with Hack4Dev. Four days, teams of four assigned by the organisers. Did not place; the spatial-leakage finding is the work worth reading.

ACES CodeFest 2026 — 2nd Place — UI/UX Design (July 2026)

- Redesigned the ACES KNUST website for mobile across all nine pages as a team of five.

Robofest Ghana — National Champion — Robotics (2025)

- Built autonomous robots under timed conditions to complete designated tasks; designed and assembled within strict time limits.

National Cybersecurity Challenge — Regional Champion (2024)

- A written challenge on the basics of cyber defence: threat analysis, network security and good practice. Won the regional round and went through to the national stage.

International Public Policy Forum — Competitor (2024-25)

- Global policy debate organised by the Brewer Foundation and New York University. Co-wrote the qualifying essay arguing that fair access to medicine should come before patent rights.

TECHNICAL WRITING

- “The scoring was lying to us” — how 82.4% of our test pixels turned out to be touching a training pixel, and what fixing it cost.
- “What transfer learning actually costs” — three assumptions about adapting a model to a new city, all measured, all wrong.
- “From PID Control to Reinforcement Learning” — what tuning Kp, Ki and Kd on a championship robot taught me about learning rates.
- “Three Bugs That Taught Me More Than Any Tutorial” — React Native zIndex, a stale closure, and nine haptic iterations.
- All four at oppong-py.github.io/writing.html

LEADERSHIP & VOLUNTEERING

Club Leader — Prempeh College Cybersecurity Club (2023–2025)

- Founded the club, ran sessions on ethical hacking, network security and CTF challenges, and personally taught 7 junior students the fundamentals.

President — House Christian Fellowship, Prempeh College (2024–2025)

REFERENCES

- Four named referees, with their own words and signed letters available on request: a robotics patron, two senior teachers from Prempeh College, and the Executive Director of SFLIG. Full details at oppong-py.github.io/testimonials.html